Kleiss-Kuijf Relations from Momentum Amplituhedron Geometry

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Positive Geometries for Scattering Amplitudes and Beyond
7-18 June 2021

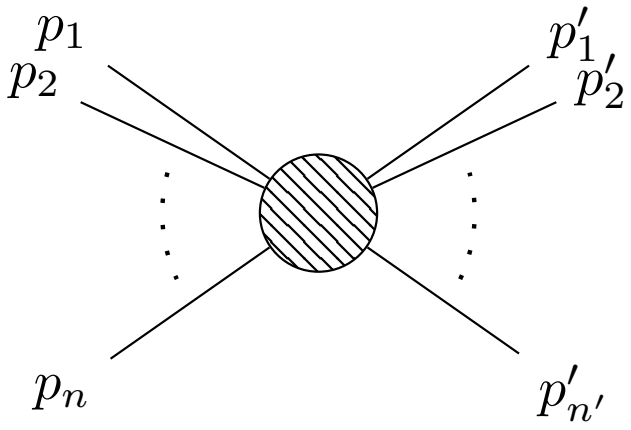
Based on [Damgaard, Ferro, Łukowski, Moerman 2103.13908].

Related work:

- $k = 2$: [Arkani-Hamed, Bourjaily, Cachazo, Postnikov, Trnka 1412.8475]

Introduction

Scattering amplitudes are *functions* of the kinematic space of *on-shell* particles living on the boundary of flat space-time.



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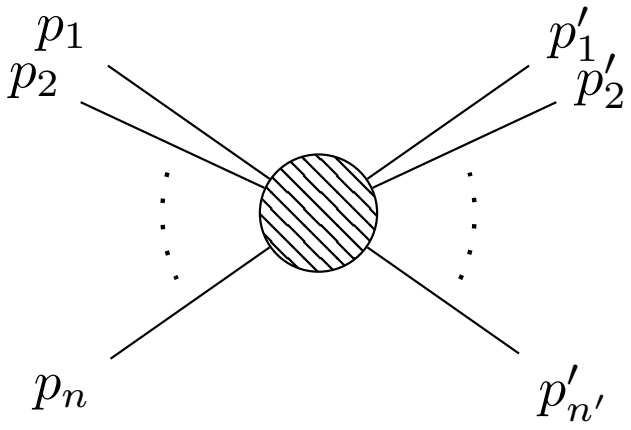

Can we obtain scattering amplitudes as the answer to a question posed directly in the kinematic space?

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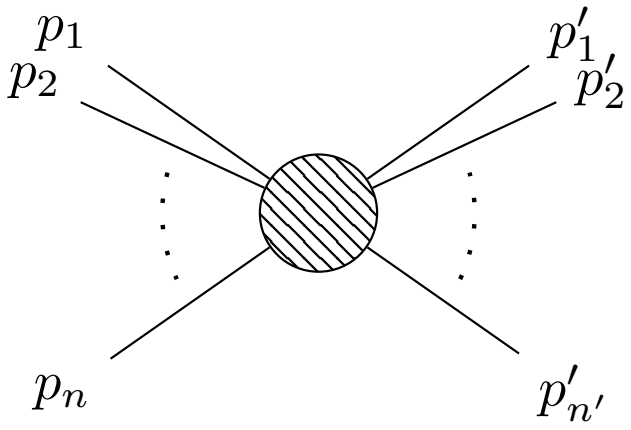

Can we obtain scattering amplitudes as the answer to a question posed directly in the kinematic space?

One perspective: positive geometries and canonical forms

Introduction

Positive Geometries

[Arkani-Hamed, Bai, Lam 1703.04541]

complex projective variety
$\dim_{\mathbb{C}}(X) = d$

$(X, X_{\geq 0})$

closed, oriented, semi-algebraic subset of $X(\mathbb{R})$
$\dim_{\mathbb{R}}(X_{\geq 0}) = d$

+ unique top-form $\Omega(X, X_{\geq 0})$ (canonical form)

- $d > 0$: each boundary component $(C, C_{\geq 0})$ is a co-dimension-one positive geometry with $\Omega(C, C_{\geq 0}) = \mathrm{Res}_C \Omega(X, X_{\geq 0})$
- $d = 0$: $X = X_{\geq 0}$ is a single point and $\Omega(X, X_{\geq 0}) = \pm 1$

Introduction

Positive Geometries

[Arkani-Hamed, Bai, Lam 1703.04541]

Positive Geometries		**Scattering Amplitudes**
recursive structure of boundaries	$\Longleftrightarrow$	factorization on physical poles

Geometrization: encode scattering amplitudes in the canonical form of a positive geometry defined in the kinematic space and cut out by "positivity conditions".

Colour Decomposition in $SU(N_c)$ Gauge Theory

1. *trace decomposition*

$$A_n^{\text{tree}} = g^{n-2} \sum_{\sigma \in S_{\{2,\ldots,n\}}} \text{Tr}(T^{a_1} T^{a_{\sigma(2)}} \ldots T^{a_{\sigma(n)}}) A_n[1,\sigma]$$

- cyclicity: $A_n[1,\ldots,n] = A_n[2,\ldots,n,1] = \ldots = A_n[n,1,\ldots,n-1]$
- reflection symmetry: $A_n[1,\ldots,n] = (-1)^n A_n[n,n-1,\ldots,1]$
- photon decoupling: $A_n[1,\ldots,n] + A_n[2,1,\ldots,n] + \ldots + A_n[2,\ldots,1,n] = 0$

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2. *structure-constant decomposition* [Del Duca, Dixon, Maltoni hep-ph/9910563]

$$A_n^{\text{tree}} = g^{n-2} \sum_{\sigma \in S_{\{2,\ldots,n-1\}}} f^{a_1 a_{\sigma(2)} b_2} f^{b_2 a_{\sigma(3)} b_3} \ldots f^{b_{n-2} a_{\sigma(n-1)} a_n} A_n[1,\sigma,n]$$

- Kleiss-Kuijf (KK) relations: [Kleiss, Kuijf 1989]

$$A_n[1,\alpha,n,\beta] = (-1)^{|\beta|} \sum_{\sigma \in \alpha \sqcup \beta^T} A_n[1,\sigma,n]$$

Colour Decomposition in $SU(N_c)$ Gauge Theory

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- Kleiss-Kuijf (KK) relations: [Kleiss, Kuijf 1989]

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Can we find a geometric origin for the Kleiss-Kuijf relations?

Amplitudes in $\mathcal{N} = 4$

$\mathcal{N} = 4$ supersymmetric Yang-Mills theory (SYM)

$\rightarrow$ prototype for QCD
$\rightarrow$ test-bed for developing new formulations for QFT

- Geometrization of amplitudes in planar $\mathcal{N} = 4$ SYM: Amplituhedron [Arkani-Hamed, Trnka 1312.2007]
- Geometrization of tree-level amplitudes in $\mathcal{N} = 4$ SYM: Momentum Amplituhedron [Damgaard, Ferro, Łukowski, Parisi 1905.04216]

Amplitudes in $\mathcal{N} = 4$ SYM

$$\lambda_i^a \tilde{\lambda}_i^{\dot{a}} = p_i^{a\dot{a}} = x_i^{a\dot{a}} - x_{i+1}^{a\dot{a}}$$
$$\lambda_i^a \eta_i^A = q_i^{aA} = \theta_i^{aA} - \theta_{i+1}^{aA}$$

on-shell superspace

$$(\lambda_i^a, \tilde{\lambda}_i^{\dot{a}} | \eta_i^A)$$
$$p^2 = 0 \iff p_i^{a\dot{a}} = \lambda_i^a \tilde{\lambda}_i^{\dot{a}}$$

$\longleftrightarrow$

dual superspace

$$(\lambda_i^a, x_i^{a\dot{a}} | \theta_i^{aA})$$

Fourier transform on λ_i^a $\updownarrow$

Incidence Relations $\updownarrow$

$$\tilde{\mu}_i^{\dot{a}} = x_i^{a\dot{a}} \lambda_{ia}$$
$$\chi_i^A = \theta_i^{aA} \lambda_{ia}$$

twistor superspace

$$\mathcal{W}_i^{\mathsf{A}} = (\mu_i^a, \tilde{\lambda}_i^{\dot{a}} | \eta_i^A)$$

momentum-twistor superspace

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Momentum Amplituhedron

Fourier transform on λ_i^a

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Amplitudes in $\mathcal{N} = 4$ SYM as Differential Forms

Key idea: uplift scattering amplitudes from *functions* to *differential forms* on the kinematic space [He, Zhang 1807.11051]

- Fourier transform to *non-chiral on-shell superspace*:

chiral on-shell superspace
$(\lambda_i^a, \tilde{\lambda}_i^{\dot{a}} \vert \eta_i^A),\ a, \dot{a} = 1, 2,\ A = 1, 2, 3, 4$

$$q^{aA} = \sum_{i=1}^n \lambda_i^a \tilde{\eta}_i^A$$

on two η's

non-chiral on-shell superspace
$(\lambda_i^a, \tilde{\lambda}_i^{\dot{a}} \vert \eta_i^r, \tilde{\eta}_i^{\dot{r}}),\ a, \dot{a}, r, \dot{r} = 1, 2$

$$\begin{cases} q^{a\dot{r}} = \sum_{i=1}^n \lambda_i^a \tilde{\eta}_i^{\dot{r}} \\ \tilde{q}^{\dot{a}r} = \sum_{i=1}^n \tilde{\lambda}_i^{\dot{a}} \eta_i^r \end{cases}$$

- associate $a, \dot{a}$ with $SU(2) \times SU(2)$ R-symmetry indices $r, \dot{r}$.
- replace Grassmann-odd parameters with *differential forms*:

$$\eta_i^a \to d\lambda_i^a\,, \qquad \tilde{\eta}_i^{\dot{a}} \to d\tilde{\lambda}_i^{\dot{a}}$$

Amplitudes in $\mathcal{N} = 4$ SYM as Differential Forms

Key idea: uplift scattering amplitudes from *functions* to *differential forms* on the kinematic space [He, Zhang 1807.11051]

n-point super-amplitude in non-chiral on-shell superspace $\leftrightarrow$ $2n$ form

$$A_{n,k}[\sigma]\Big|_{\eta_i^a \to d\lambda_i^a, \tilde{\eta}_i^{\dot{a}} \to d\tilde{\lambda}_i^{\dot{a}}} = d^4 q \wedge \boldsymbol{\Omega}_{n,k}^{(\sigma)}, \quad \begin{cases} q^{a\dot{a}} = \sum_{i=1}^n \lambda_i^a d\tilde{\lambda}_i^{\dot{a}} \\ \tilde{q}^{\dot{a}r} = \sum_{i=1}^n \tilde{\lambda}_i^{\dot{a}} d\lambda_i^a \end{cases}$$

$2n-4$ form

Momentum Amplituhedron $\mathcal{M}_{n,k}^{(\sigma)}$

positive geometry with canonical form $\boldsymbol{\Omega}_{n,k}^{(\sigma)}$

Momentum Amplituhedron

[Damgaard, Ferro, Łukowski, Parisi 1905.04216]

Momentum Amplituhedron $\mathcal{M}_{n,k}^{(\sigma)}$

positive geometry with canonical form $\boldsymbol{\Omega}_{n,k}^{(\sigma)}$

- encodes *tree-level*, *colour-ordered* amplitudes in $\mathcal{N} = 4$ SYM
- defined on *on-shell space* $(\lambda_i^a, \tilde{\lambda}_i^{\dot{a}}) : \sum_{i=1}^n \lambda_i^a \tilde{\lambda}_i^{\dot{a}} = 0$
- closed, oriented region satisfying "positivity conditions"

Two definitions:

1. using auxiliary space
2. directly in on-shell space

Momentum Amplituhedron

1 using auxiliary space

- $2n$ Grassmann-odd auxiliary variables $(\tilde{\phi}^{\dot{\alpha}}_{\dot{a}}, \phi^{\alpha}_{a})$
  $a, \dot{a} = 1, 2,\ \dot{\alpha} = 1, \ldots, k,\ \alpha = 1, \ldots, n-k.$
- external data: *bosonized* non-chiral on-shell superspace variables

positive $(k+2) \times n$ matrix

$$\tilde{\Lambda}^{\dot{A}}_{i} = \begin{pmatrix} \tilde{\lambda}^{\dot{a}}_{i} \\ \tilde{\phi}^{\dot{\alpha}}_{a} \tilde{\eta}^{\dot{a}}_{i} \end{pmatrix} \qquad \Lambda^{A}_{i} = \begin{pmatrix} \lambda^{a}_{i} \\ \phi^{\alpha}_{a} \eta^{a}_{i} \end{pmatrix}$$

twisted positive $(n-k+2) \times n$ matrix

- *Momentum Amplituhedron:* image of the *totally non-negative Grassmannian* $G_{\geq 0}(k,n)$ via the linear map

$$\Phi_{(\tilde{\Lambda},\Lambda)} : G_{\geq 0}(k,n) \to G(k,k+2) \times G(n-k,n-k+2)$$

$$\tilde{Y}^{\dot{A}}_{\dot{\alpha}} = \sum_{i=1}^{n} c_{\dot{\alpha} i} \tilde{\Lambda}^{\dot{A}}_{i} \qquad Y^{A}_{\alpha} = \sum_{i=1}^{n} c^{\perp}_{\alpha i} \Lambda^{A}_{i}$$

Momentum Amplituhedron

1 using auxiliary space

$$\mathcal{M}_{n,k} \equiv \Phi_{(\tilde{\Lambda},\Lambda)}(G_{\geq 0}(k,n))$$

- *Volume form:* for any *positroidal triangulation* $\{c^{(\pi)}\}$ of $\mathcal{M}_{n,k}$

$$\boldsymbol{\Omega}_{n,k} = \Phi_{(\tilde{\Lambda},\Lambda)*} \sum_{\pi} d\log \alpha_1^{(\pi)} \wedge \cdots \wedge d\log \alpha_{2n-4}^{(\pi)}$$

- *Volume function:* $P^{a\dot{a}} = \sum_{i=1}^{n} (Y^{\perp} \cdot \Lambda)_i^a (\tilde{Y}^{\perp} \cdot \tilde{\Lambda})_i^{\dot{a}}$

$$\boldsymbol{\Omega}_{n,k} \wedge d^4P\, \delta^{(4)}(P) = \prod_{\alpha=1}^{n-k} \langle Y_1 \ldots Y_{n-k}\, d^2Y_\alpha \rangle \prod_{\dot{\alpha}=1}^{k} [\tilde{Y}_1 \ldots \tilde{Y}_k\, d^2\tilde{Y}_{\dot{\alpha}}]\, \delta^{(4)}(P)\, \Omega_{n,k}$$

- *Amplitude:* reference subspaces $Y^* = \begin{pmatrix} 0_{2\times(n-k)} \\ \mathbb{1}_{(n-k)\times(n-k)} \end{pmatrix}$ and $\tilde{Y}^* = \begin{pmatrix} 0_{2\times k} \\ \mathbb{1}_{k\times k} \end{pmatrix}$

$$A_{n,k} = \delta^{(4)}(p) \int d\phi_a^1 \ldots d\phi_a^{n-k} \int d\tilde{\phi}_{\dot{a}}^1 \ldots d\tilde{\phi}_{\dot{a}}^k\, \Omega_{n,k}(Y^*, \tilde{Y}^*, \Lambda, \tilde{\Lambda})$$

Momentum Amplituhedron

1 using auxiliary space

$$\mathcal{M}_{n,k} \equiv \Phi_{(\tilde{\Lambda},\Lambda)}(G_{\geq 0}(k,n))$$

Facets: co-dimension-one boundaries

- collinear limits: $\langle Y\, i\, i+1\rangle = 0$ and $[\tilde{Y}\, i\, i+1] = 0$
- multi-particle factorizations: $S_{i,i+1,\ldots,i+p} = 0$ for $p = 2, \ldots, n-4$

$$S_{i,i+1,\ldots,i+p} = \sum_{i \leq j_1 < j_2 \leq i+p} \langle Y\, j_1\, j_2\rangle [\tilde{Y}\, j_1\, j_2]$$

$\mathcal{M}_{n,k}$ inherits its boundary stratification from $G_{\geq 0}(k,n)$

- `positroids.m` [Bourjaily 1212.6974] for $G_{\geq 0}(k,n)$
- `amplituhedronBoundaries.m` [Łukowski, Moerman 2002.07146] for $\mathcal{M}_{n,k}$

Physical Singularities and Geometry

Geometrization: boundaries of $\mathcal{M}_{n,k}$ encode all physical singularities of *tree-level, colour-ordered* amplitudes in $\mathcal{N}=4$ SYM [Ferro, Łukowski, Moerman 2003.13704]

- multi-particle factorizations: $A_{n,k}(1,\ldots,n) \xrightarrow{P_{i,j}^2 = s_{i,i+1,\ldots,j} \to 0}$

$$\sum_{k'=1}^{k} \int d^4\eta_{P_{i,j}} A_{j-i+2,k'}(i,\ldots,j,P_{i,j}) \frac{1}{P_{i,j}^2} A_{n-j+i,k-k'+1}(P_{i,j},j+1,\ldots,i-1)$$

- collinear limits: $A_{n,k}(1,\ldots,n) \xrightarrow{p_1 || p_2}$

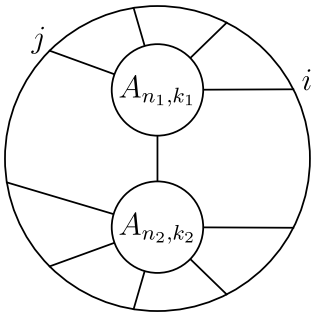
$$\int d^4\eta_{P_{i,j}} [\underset{\langle 12\rangle \to 0}{\mathrm{Split}_0^{\mathrm{tree}}} A_{n-1,k}(P_{1,2},3,\ldots,n) + \underset{[12]\to 0}{\mathrm{Split}_{-1}^{\mathrm{tree}}} A_{n-1,k-1}(P_{1,2},3,\ldots,n)]$$

- soft limits: $A_{n,k}(1,\ldots,n) \xrightarrow{p_2 \to 0}$

$$\underset{\langle 12\rangle,\langle 23\rangle \to 0}{\mathrm{Soft}_0^{\mathrm{tree}}} A_{n-1,k}(1,3,\ldots,n) + \underset{[12],[23]\to 0}{\mathrm{Soft}_{-1}^{\mathrm{tree}}} A_{n-1,k-1}(1,3,\ldots,n)$$

Physical Singularities and Geometry

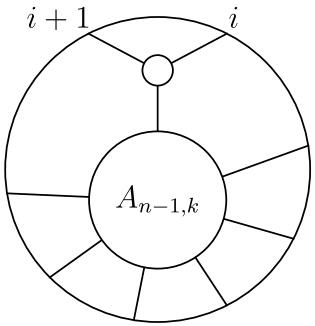
- multi-particle factorizations:



$$s_{i,i+1,\ldots,j} = 0$$

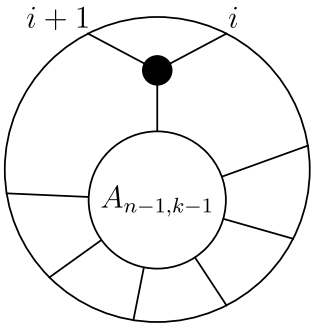
- collinear limits:

$\mathrm{Split}_0^{\mathrm{tree}}$



$$\langle i\, i+1\rangle = 0$$

$\mathrm{Split}_{-1}^{\mathrm{tree}}$



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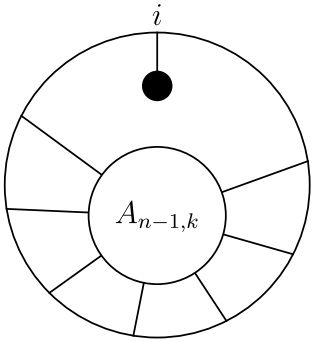
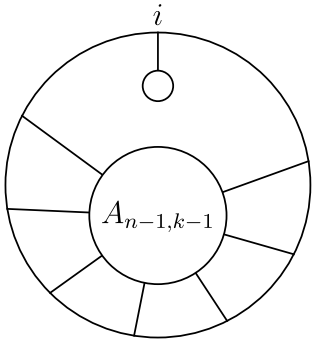


$$\langle i-1\, i\rangle = 0$$
$$\langle i\, i+1\rangle = 0$$

$\mathrm{Soft}_{-1}^{\mathrm{tree}}$



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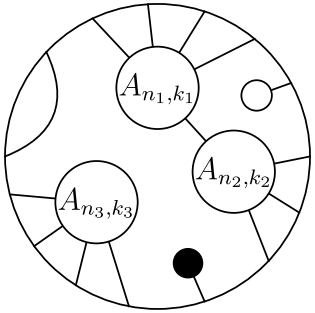
- generic limit:



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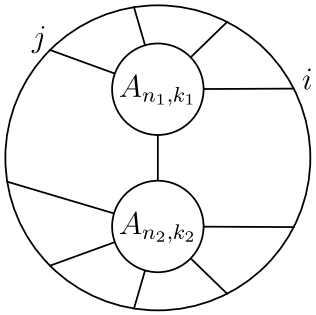
Geometrization: boundaries of $\mathcal{M}_{n,k}$ encode all physical singularities of *tree-level, colour-ordered* amplitudes in $\mathcal{N} = 4$ SYM [Ferro, Łukowski, Moerman 2003.13704]

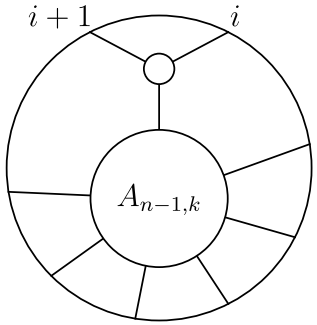
Co-dimension-one Boundaries

factorization limit



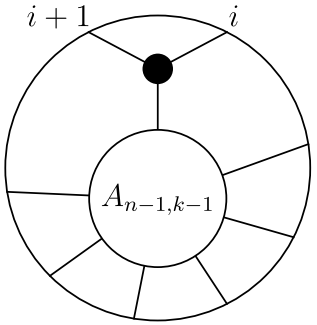
$$S_{i,i+1,\cdots,j} = 0$$

k-preserving collinear limit



$$\langle Y i i+1 \rangle = 0$$

k-reducing collinear limit



$$[\tilde{Y} i i+1] = 0$$

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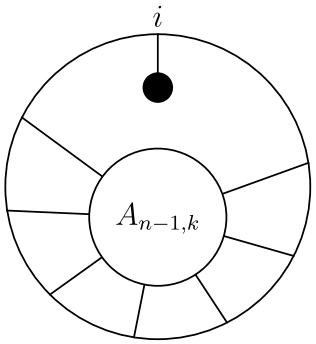
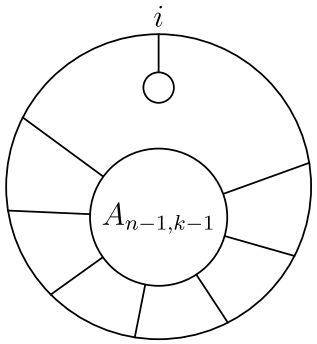
Co-dimension-two Boundaries

k-preserving soft limit



$$\langle Y i-1 i\rangle = 0$$
$$\langle Y i i+1\rangle = 0$$

k-reducing soft limit



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$+$ further factorization and collinear limits

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Higher Co-dimension Boundaries

obtained by intersecting co-dimension-one boundaries

Can be labelled by[1]

$$\begin{array}{lcl} \langle Y i i+1 \rangle = 0 & \Longleftrightarrow & k\text{-preserving collinear limit} \\ [\tilde{Y} i i+1] = 0 & \Longleftrightarrow & k\text{-reducing collinear limit} \\ S_{i,i+1,\ldots,i+p} = 0 & \Longleftrightarrow & \text{multi-particle Factorization limit} \\ P_{i,i+1,\ldots,i+p} = 0 & \Longleftrightarrow & \text{multi-particle soft limit} \end{array}$$

[1] Important for homological algorithm on slide 42.

Momentum Amplituhedron

2 directly in on-shell space
(standard ordering)

- define a $(2n-4)$-dimensional affine subspace of on-shell space:

$$\mathcal{V}_{n,k} \equiv \{(\lambda_i^a, \tilde{\lambda}_i^{\dot{a}}) : \lambda_i^a = \lambda_i^{*a} + y_\alpha^a \Delta_i^\alpha,\ \tilde{\lambda}_i^{\dot{a}} = \tilde{\lambda}_i^{*\dot{a}} + \tilde{y}_{\dot{\alpha}}^{\dot{a}} \tilde{\Delta}_i^{\dot{\alpha}},\ \sum_{i=1}^n \lambda_i^a \tilde{\lambda}_i^{\dot{a}} = 0\}$$

with $(\lambda^*, \tilde{\lambda}^*)$ two fixed 2-planes, $\tilde{\Delta}$ fixed k-plane, Δ fixed $(n-k)$-plane in n dimensions

positive $(k+2) \times n$ matrix

$$\tilde{\Lambda}_i^{\dot{A}} = \begin{pmatrix} \tilde{\lambda}_i^{*\dot{a}} \\ \tilde{\Delta}_i^{\dot{\alpha}} \end{pmatrix} \qquad \Lambda_i^A = \begin{pmatrix} \lambda_i^{*a} \\ \Delta_i^\alpha \end{pmatrix}$$

twisted positive $(n-k+2) \times n$ matrix

$(y_\alpha^a, \tilde{y}_{\dot{\alpha}}^{\dot{a}})$ + momentum conservation $\implies 2(n-k)+2k-4 = 2n-4$ variables

Momentum Amplituhedron

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- define a $(2n-4)$-dimensional affine subspace of on-shell space:

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- define a *"winding space"*:

$$\mathcal{W}_{n,k} \equiv \{(\lambda_i^a, \tilde{\lambda}_i^{\dot{a}}) : \langle ii+1\rangle \geq 0,\ [ii+1] \geq 0,\ s_{i,i+1,\ldots,i+p} \geq 0,$$
$$\text{sequence } \{\langle 12\rangle, \langle 13\rangle, \ldots \langle 1n\rangle\} \text{ has } k-2 \text{ sign flips}$$
$$\text{sequence } \{[12], [13], \ldots [1n]\} \text{ has } k \text{ sign flips}\}$$

- *Momentum Amplituhedron:*

$$\mathcal{M}_{n,k} \equiv \mathcal{V}_{n,k} \cap \mathcal{W}_{n,k}$$

- *Amplitude:* $A_{n,k}[1,\ldots,n] = \delta^{(4)}(p) d^4p \wedge \mathbf{\Omega}_{n,k}\Big|_{d\lambda\to\eta, d\tilde{\lambda}\to\tilde{\eta}}$

Momentum Amplituhedron

2 directly in on-shell space
(non-standard ordering σ)

[Damgaard, Ferro, Łukowski, Moerman 2103.13908]

- same $(2n-4)$-dimensional affine subspace of on-shell space:

$$\mathcal{V}_{n,k} \equiv \{(\lambda_i^a, \tilde{\lambda}_i^{\dot{a}}) : \lambda_i^a = \lambda_i^{*a} + y_\alpha^a \Delta_i^\alpha,\ \tilde{\lambda}_i^{\dot{a}} = \tilde{\lambda}_i^{*\dot{a}} + \tilde{y}_{\dot{\alpha}}^{\dot{a}} \tilde{\Delta}_i^{\dot{\alpha}},\ \sum_{i=1}^n \lambda_i^a \tilde{\lambda}_i^{\dot{a}} = 0\}$$

- define a *"winding space"*: $\mathcal{W}_{n,k}^{(\sigma)} \equiv \{(\lambda_i^a, \tilde{\lambda}_i^{\dot{a}}) : (\lambda_{\sigma(i)}^a, \tilde{\lambda}_{\sigma(i)}^{\dot{a}}) \in \mathcal{W}_{n,k}\}$
- *Momentum Amplituhedron:*

$$\mathcal{M}_{n,k}^{(\sigma)} \equiv \mathcal{V}_{n,k} \cap \mathcal{W}_{n,k}^{(\sigma)}, \qquad \mathbf{\Omega}_{n,k}^{(\sigma)} = \mathbf{\Omega}_{n,k}\Big|_{\lambda_i \to \lambda_{\sigma(i)}, \tilde{\lambda}_i \to \tilde{\lambda}_{\sigma(i)}}$$

- *Amplitude:* $A_{n,k}[\sigma] = \delta^{(4)}(p) d^4 p \wedge \mathbf{\Omega}_{n,k}^{(\sigma)}\Big|_{d\lambda \to \eta, d\tilde{\lambda} \to \tilde{\eta}}$

Kleiss-Kuijf Relations from Positive Geometry

[Damgaard, Ferro, Łukowski, Moerman 2103.13908]

Can we find a geometric origin for the Kleiss-Kuijf relations?

Kleiss-Kuijf Relations from Positive Geometry

[Damgaard, Ferro, Łukowski, Moerman 2103.13908]

Can we find a geometric origin for the Kleiss-Kuijf relations?

$$\sum_i c_i \, A_{n,k}[\sigma_i] = 0 \iff \sum_i c_i \, \Omega_{n,k}^{(\sigma_i)} = 0 \,, \qquad (c_i = \pm 1)$$

Kleiss-Kuijf Relations from Positive Geometry

[Damgaard, Ferro, Łukowski, Moerman 2103.13908]

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Can we determine when a collection of canonical forms will sum to zero?

Kleiss-Kuijf Relations from Positive Geometry

[Damgaard, Ferro, Łukowski, Moerman 2103.13908]

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Can we determine when a collection of canonical forms will sum to zero?

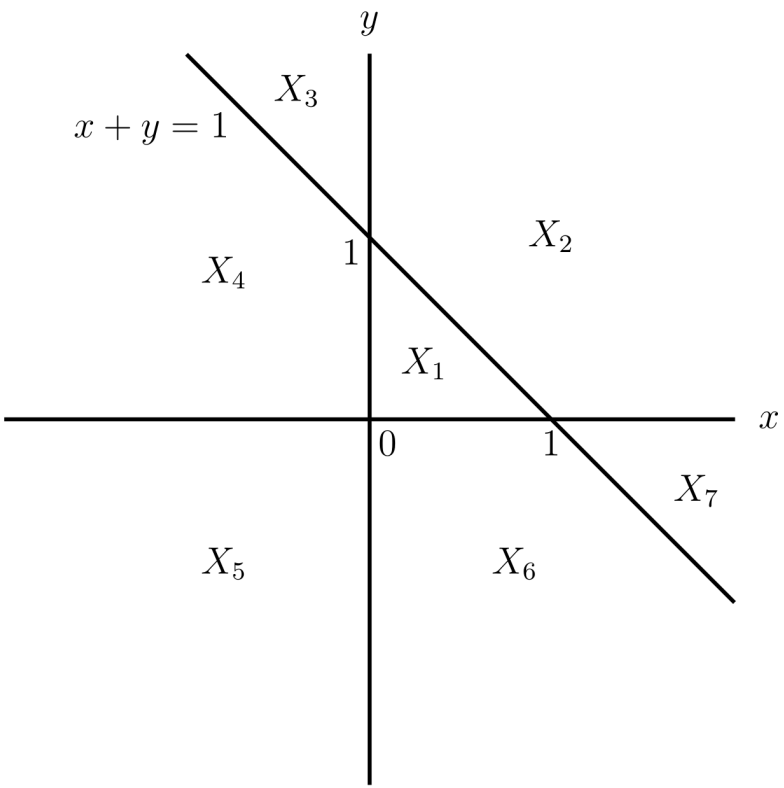
Oriented sum: an operation on positive geometries which captures the addition of canonical forms.

For positive geometries X_1 and X_2, define the oriented sum $X_1 \oplus X_2$ such that $\Omega(X_1 \oplus X_2) = \Omega(X_1) + \Omega(X_2)$.

Kleiss-Kuijf Relations from Positive Geometry

[Damgaard, Ferro, Łukowski, Moerman 2103.13908]

Oriented Sums

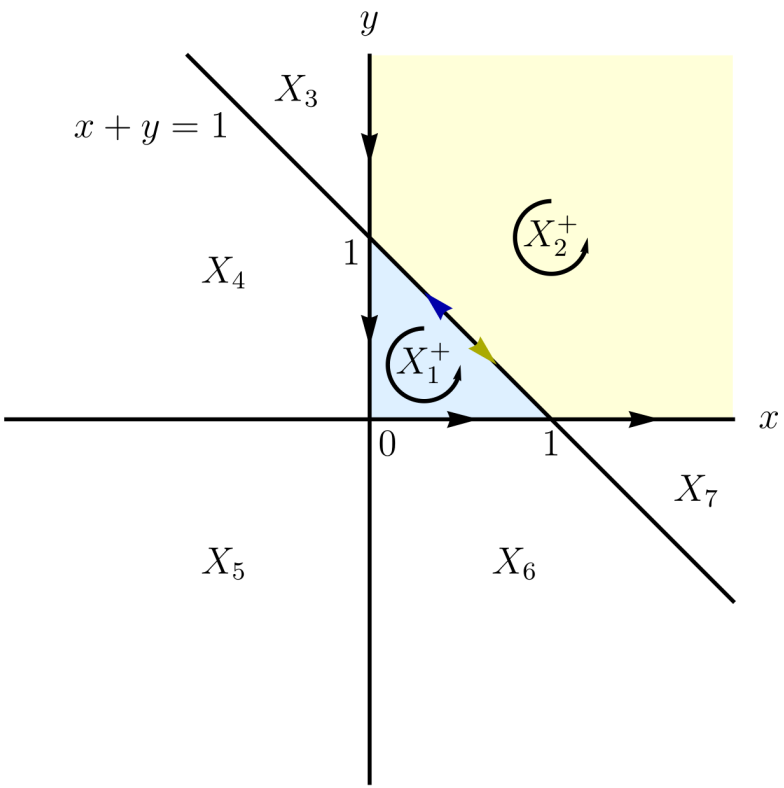


- $2d$ plane decomposed into 7 regions
- $X_8 \equiv \mathbb{R}^2_{\geq 0}$ and $X_9 \equiv \mathbb{R}_{\leq 0} \times \mathbb{R}_{\geq 0}$
- X_i^+ or X_i^- = anti-clockwise or clockwise
- $\Omega_i^\pm = \Omega(X_i^\pm)$ and $\Omega_i^- = -\Omega_i^+$

Kleiss-Kuijf Relations from Positive Geometry

[Damgaard, Ferro, Łukowski, Moerman 2103.13908]

Oriented Sums



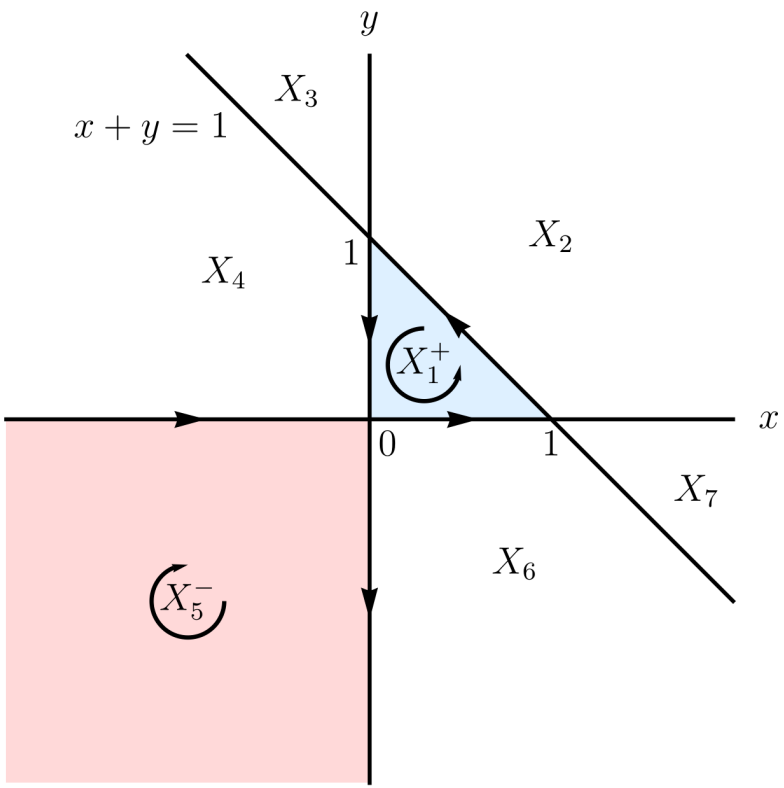
Example 1: $X_1^+ \oplus X_2^+ = X_8^+$

- positive geometry with $\Omega = d\log x \wedge d\log y$
- akin to an *internal triangulation*

Kleiss-Kuijf Relations from Positive Geometry

[Damgaard, Ferro, Łukowski, Moerman 2103.13908]

Oriented Sums



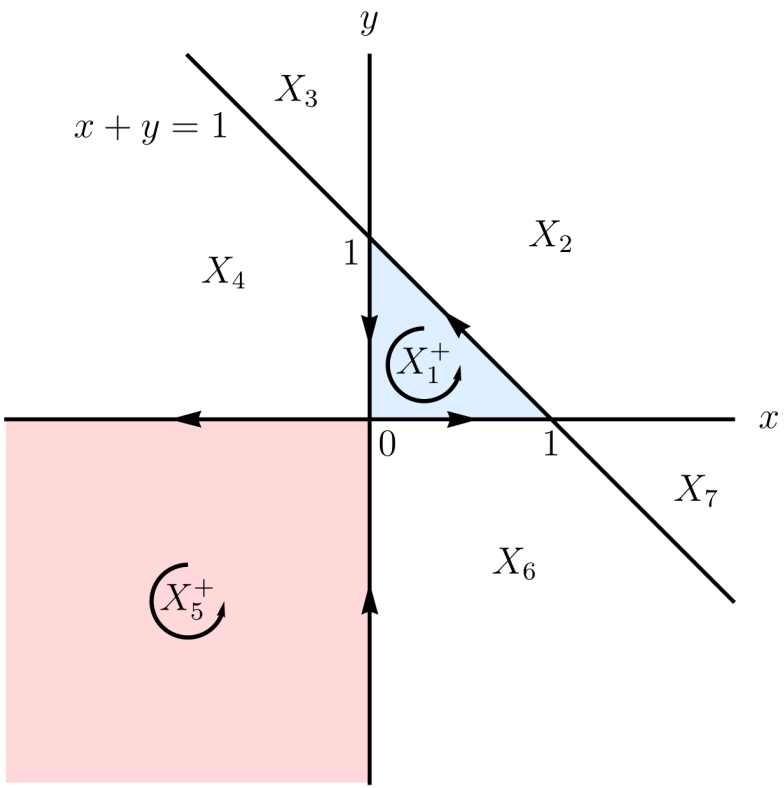
Example 2: $X_1^+ \oplus X_5^- \cong X_2^-$

- positive geometry with
  $\Omega = d\log y \wedge d\log(x+y-1)$
  $+ d\log(x+y-1) \wedge d\log x = \Omega_2^-$
- vertex at $(0,0)$ *disappears* in oriented sum

Kleiss-Kuijf Relations from Positive Geometry

[Damgaard, Ferro, Łukowski, Moerman 2103.13908]

Oriented Sums



Example 3: $X_1^+ \oplus X_5^+$

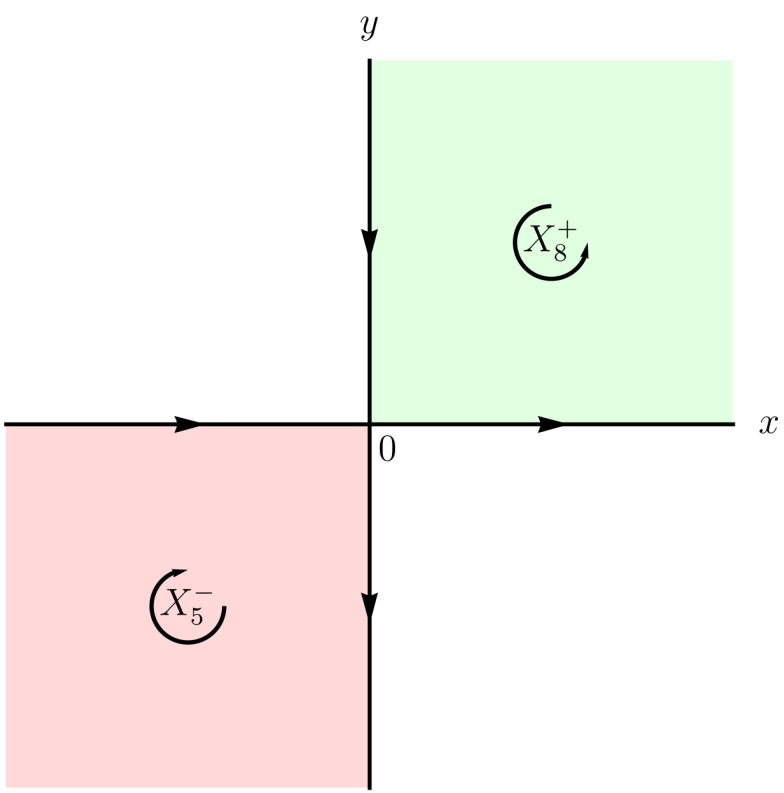
- residues at $(0,0)$ are ± 2
- residues at $(1,0)$ and $(0,1)$ are ± 1
- does not have uniform leading singularities

  $\Longrightarrow$ *not* a positive geometry

Kleiss-Kuijf Relations from Positive Geometry

[Damgaard, Ferro, Łukowski, Moerman 2103.13908]

Oriented Sums



Example 4: $X_8^+ \oplus X_5^-$

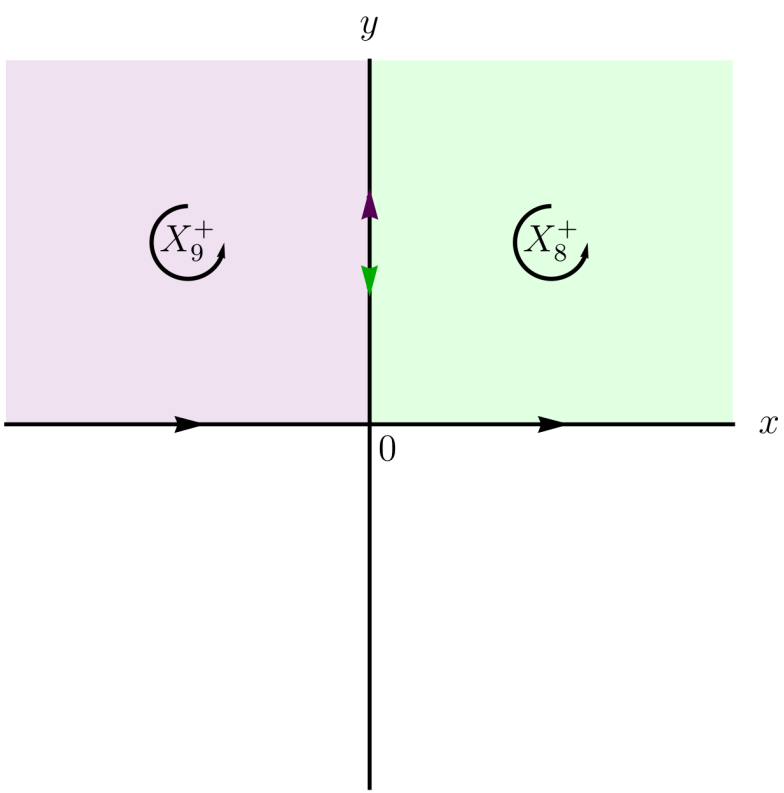
- $\Omega_8^+ + \Omega_5^- = 0$
- vertex at $(0,0)$ *disappears* in oriented sum
- does not have boundaries of all co-dimensions

  $\Longrightarrow$ *not* a positive geometry

Kleiss-Kuijf Relations from Positive Geometry

[Damgaard, Ferro, Łukowski, Moerman 2103.13908]

Oriented Sums



Example 5: $X_8^+ \oplus X_9^+$

- $\Omega_8^+ + \Omega_9^+ = 0$
- upper half plane
- vertex at $(0,0)$ *disappears* in oriented sum
- does not have boundaries of all co-dimensions
  $\Longrightarrow$ *not* a positive geometry

Kleiss-Kuijf Relations from Positive Geometry

[Damgaard, Ferro, Łukowski, Moerman 2103.13908]

Can we determine when a collection of canonical forms will sum to zero?

Claim: a collection of canonical forms will sum to zero *iff* the oriented sum *has no vertices.*

$\Longrightarrow$ we can determine the Kleiss-Kuijf relations *directly* from geometry (without needing the canonical form)

Kleiss-Kuijf Relations from Positive Geometry

[Damgaard, Ferro, Łukowski, Moerman 2103.13908]

MHV Amplitudes: Simplicial Realization

- $\mathcal{V}_{n,2}$: $(2n-4)$ y's and 4 $\tilde{y}$'s + momentum conservation
  $\implies$ fix 4 $\tilde{y}$'s and λ has $2n-4$ degrees of freedom
- $\mathcal{W}_{n,2}$: $\langle ij \rangle \geq 0$ for all $1 \leq i < j \leq n$
- $\mathcal{W}_{n,2}^{(\sigma)}$: $\langle \sigma(i)\sigma(j) \rangle \geq 0$ for all $1 \leq i < j \leq n$
- choose patch of $\mathcal{V}_{n,2}$ where $\langle 12 \rangle \neq 0$

$$\mathcal{V}_{n,2} : \lambda = \begin{pmatrix} 1 & \sum_{i=1}^{n-2} \alpha_{2i} & \left(\sum_{i=1}^{n-3} \alpha_{2i}\right) \alpha_{2(n-2)-1} & \left(\sum_{i=1}^{n-4} \alpha_{2i}\right) \alpha_{2(n-3)-1} & \cdots & \alpha_2\alpha_3 & 0 \\ 0 & 1 & \alpha_{2(n-2)-1} & \alpha_{2(n-3)-1} & \cdots & \alpha_3 & \alpha_1 \end{pmatrix}$$

Kleiss-Kuijf Relations from Positive Geometry

[Damgaard, Ferro, Łukowski, Moerman 2103.13908]

MHV Amplitudes: Simplicial Realization

$$\mathcal{V}_{n,2} : \lambda = \begin{pmatrix} 1 & \sum_{i=1}^{n-2} \alpha_{2i} & \left(\sum_{i=1}^{n-3} \alpha_{2i}\right) \alpha_{2(n-2)-1} & \left(\sum_{i=1}^{n-4} \alpha_{2i}\right) \alpha_{2(n-3)-1} & \cdots & \alpha_2\alpha_3 & 0 \\ 0 & 1 & \alpha_{2(n-2)-1} & \alpha_{2(n-3)-1} & \cdots & \alpha_3 & \alpha_1 \end{pmatrix}$$

- use cyclicity to fix the position of particle 1 to the first entry
- $\mathcal{W}_{n,2}^{(\sigma)}$: $\langle 1\sigma(j)\rangle \geq 0$ for all $1 < j \leq n \implies$ odd α's always *non-negative*
- remaining even α's parametrize $\mathbb{R}^{n-2}$

Kleiss-Kuijf Relations from Positive Geometry

[Damgaard, Ferro, Łukowski, Moerman 2103.13908]

MHV Amplitudes: Simplicial Realization

$$\mathcal{V}_{n,2} : \lambda = \begin{pmatrix} 1 & \sum_{i=1}^{n-2} \alpha_{2i} & \left(\sum_{i=1}^{n-3} \alpha_{2i}\right) \alpha_{2(n-2)-1} & \left(\sum_{i=1}^{n-4} \alpha_{2i}\right) \alpha_{2(n-3)-1} & \cdots & \alpha_2\alpha_3 & 0 \\ 0 & 1 & \alpha_{2(n-2)-1} & \alpha_{2(n-3)-1} & \cdots & \alpha_3 & \alpha_1 \end{pmatrix}$$

$$\langle ij \rangle = 0 \iff \textstyle\sum_{l=n-j+1}^{n-i} \alpha_{2l} = 0 \text{ for } 1 < i < j \leq n$$

- defines $\binom{n-1}{2}$ co-dimension-one hyperplanes in $\mathbb{R}^{n-2}$ passing through O
- defines a *complete fan* in $\mathbb{R}^{n-2}$ with $(n-1)!$ chambers
- fan is *dual* to the permutohedron of order $(n-1)$
- chambers are *(oriented) simplicial cones* spanned by $(n-2)$ rays and having $(n-2)$ hyperplane facets: positive sectors $c[\sigma]$

Kleiss-Kuijf Relations from Positive Geometry

[Damgaard, Ferro, Łukowski, Moerman 2103.13908]

Example: MHV_4

$$\mathcal{V}_{4,2} : \lambda = \begin{pmatrix} 1 & \alpha_2 + \alpha_4 & \alpha_2\alpha_3 & 0 \\ 0 & 1 & \alpha_3 & \alpha_1 \end{pmatrix}$$

- $\mathcal{W}_{4,2}^{(\sigma)}$: $\langle \sigma(i)\sigma(j) \rangle \geq 0$ for all $1 \leq i < j \leq 4$

	(1234)	(1243)	(1324)	(1342)	(1423)	(1432)
$\langle 12 \rangle = 1$	$+$	$+$	$+$	$+$	$+$	$+$
$\langle 13 \rangle = \alpha_3$	$+$	$+$	$+$	$+$	$+$	$+$
$\langle 14 \rangle = \alpha_1$	$+$	$+$	$+$	$+$	$+$	$+$
$\langle 23 \rangle = \alpha_3\alpha_4$	$+$	$+$	$-$	$-$	$+$	$-$
$\langle 24 \rangle = \alpha_1(\alpha_2 + \alpha_4)$	$+$	$+$	$+$	$-$	$-$	$-$
$\langle 34 \rangle = \alpha_1\alpha_2\alpha_3$	$+$	$-$	$+$	$+$	$-$	$-$

- $\mathcal{W}_{4,2}^{(\sigma)}$: $\langle 1\sigma(j) \rangle \geq 0$ for all $1 < j \leq 4 \implies$ odd α's always *non-negative*

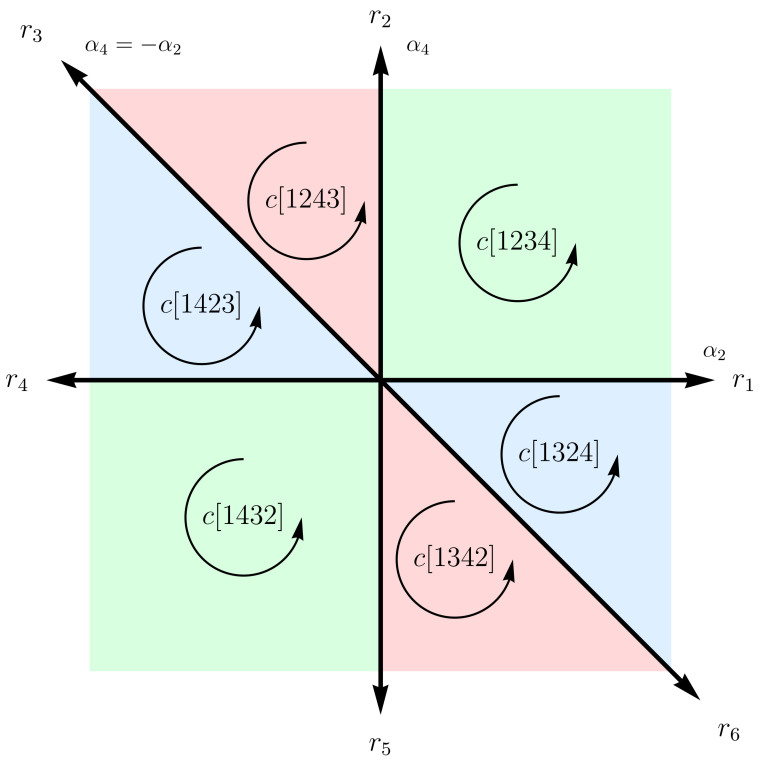
	(1234)	(1243)	(1324)	(1342)	(1423)	(1432)
$\langle 23 \rangle \sim \alpha_4$	$+$	$+$	$-$	$-$	$+$	$-$
$\langle 24 \rangle \sim \alpha_2 + \alpha_4$	$+$	$+$	$+$	$-$	$-$	$-$
$\langle 34 \rangle \sim \alpha_2$	$+$	$-$	$+$	$+$	$-$	$-$

Kleiss-Kuijf Relations from Positive Geometry

[Damgaard, Ferro, Łukowski, Moerman 2103.13908]

Example: MHV_4

	(1234)	(1243)	(1324)	(1342)	(1423)	(1432)
$\langle 23\rangle \sim \alpha_4$	$+$	$+$	$-$	$-$	$+$	$-$
$\langle 24\rangle \sim \alpha_2+\alpha_4$	$+$	$+$	$+$	$-$	$-$	$-$
$\langle 34\rangle \sim \alpha_2$	$+$	$-$	$+$	$+$	$-$	$-$



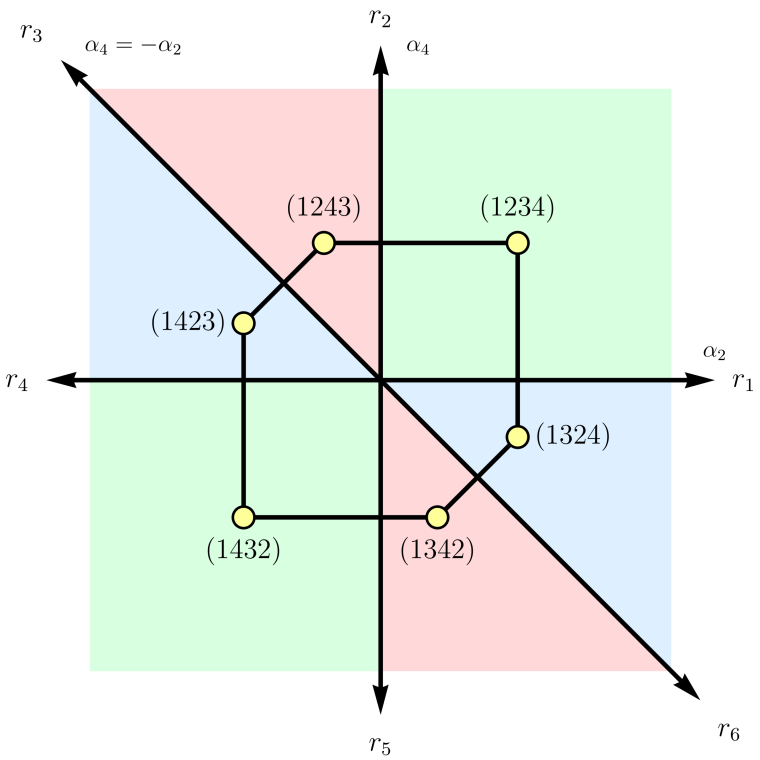
- each positive sector $c[\sigma]$ is an oriented simplicial cone spanned by 2 rays
- complete fan in $\mathbb{R}^2$

Kleiss-Kuijf Relations from Positive Geometry

[Damgaard, Ferro, Łukowski, Moerman 2103.13908]

Example: MHV_4

	(1234)	(1243)	(1324)	(1342)	(1423)	(1432)
$\langle 23\rangle \sim \alpha_4$	$+$	$+$	$-$	$-$	$+$	$-$
$\langle 24\rangle \sim \alpha_2+\alpha_4$	$+$	$+$	$+$	$-$	$-$	$-$
$\langle 34\rangle \sim \alpha_2$	$+$	$-$	$+$	$+$	$-$	$-$



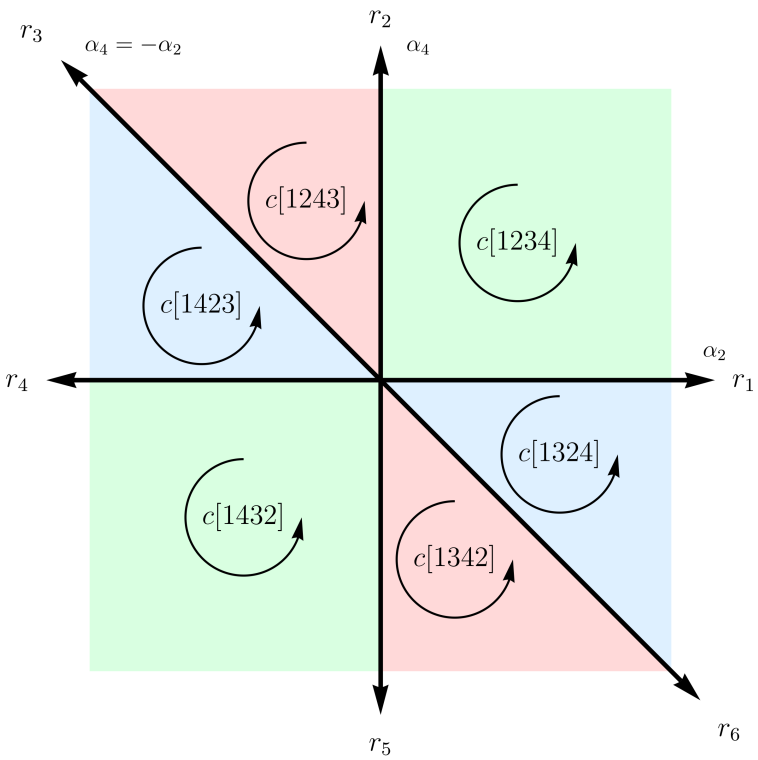
- each positive sector $c[\sigma]$ is an oriented simplicial cone spanned by 2 rays
- complete fan in $\mathbb{R}^2$ dual to the permutohedron of order 3

Kleiss-Kuijf Relations from Positive Geometry

[Damgaard, Ferro, Łukowski, Moerman 2103.13908]

Example: MHV_4

	(1234)	(1243)	(1324)	(1342)	(1423)	(1432)
$\langle 23\rangle \sim \alpha_4$	$+$	$+$	$-$	$-$	$+$	$-$
$\langle 24\rangle \sim \alpha_2+\alpha_4$	$+$	$+$	$+$	$-$	$-$	$-$
$\langle 34\rangle \sim \alpha_2$	$+$	$-$	$+$	$+$	$-$	$-$



Reflection symmetry relations:

$$A[1432] = A[1234]$$
$$A[1423] = A[1324]$$
$$A[1243] = A[1342]$$

equivalent positive geometries with *identical* canonical forms

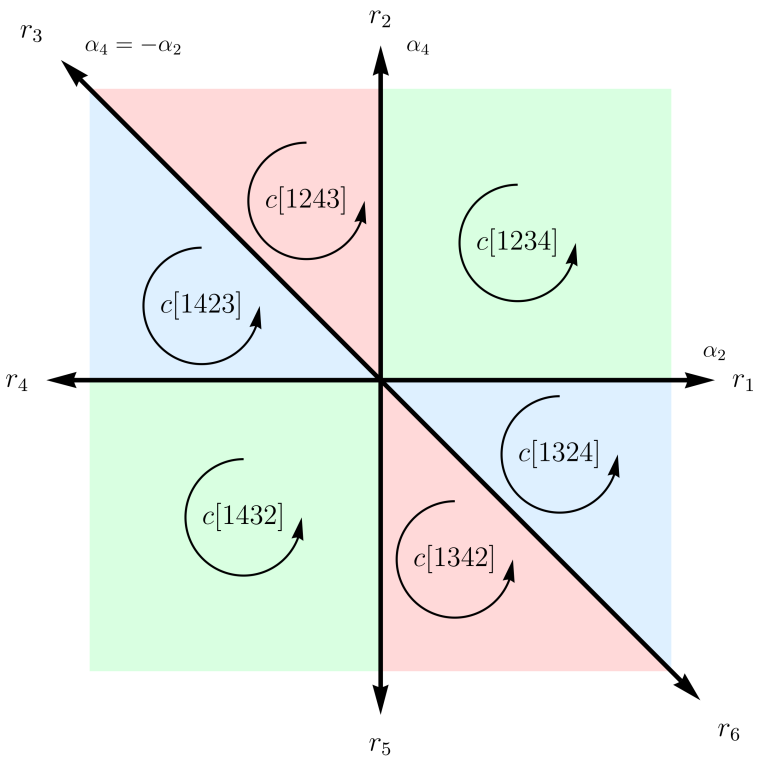
$$\Omega(c[1432]) = \Omega(c[1234]) = d\log(\alpha_4)\wedge d\log(\alpha_2)$$
$$\Omega(c[1423]) = \Omega(c[1324]) = d\log(\alpha_2+\alpha_4)\wedge d\log(\alpha_4)$$
$$\Omega(c[1243]) = \Omega(c[1342]) = d\log(\alpha_2)\wedge d\log(\alpha_2+\alpha_4)$$

Kleiss-Kuijf Relations from Positive Geometry

[Damgaard, Ferro, Łukowski, Moerman 2103.13908]

Example: MHV_4

	(1234)	(1243)	(1324)	(1342)	(1423)	(1432)
$\langle 23\rangle \sim \alpha_4$	$+$	$+$	$-$	$-$	$+$	$-$
$\langle 24\rangle \sim \alpha_2+\alpha_4$	$+$	$+$	$+$	$-$	$-$	$-$
$\langle 34\rangle \sim \alpha_2$	$+$	$-$	$+$	$+$	$-$	$-$



Reflection symmetry relations:

$$A[1432] = A[1234]$$
$$A[1423] = A[1324]$$
$$A[1243] = A[1342]$$

equivalent positive geometries with *identical* canonical forms

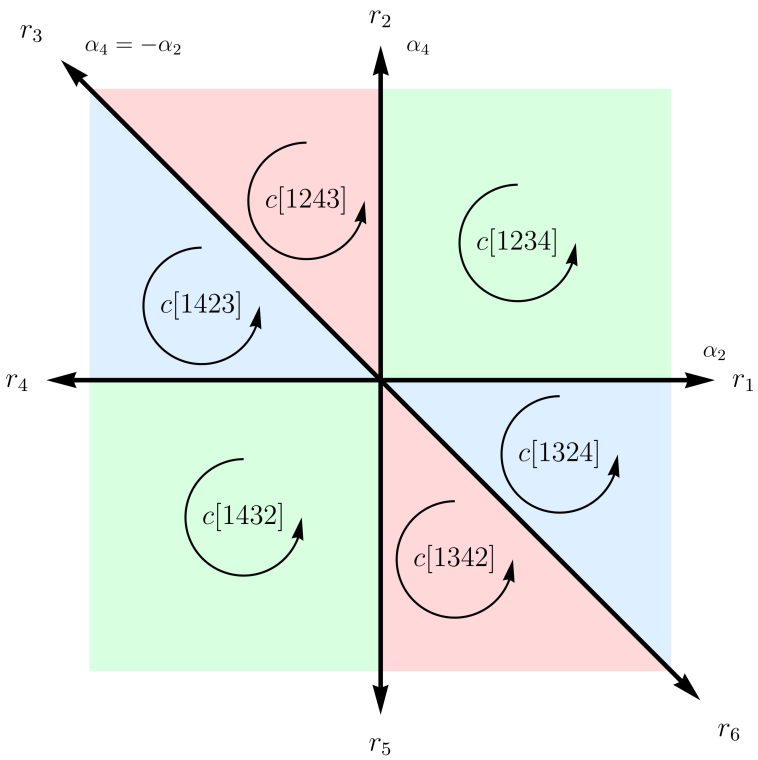
$$\Omega(c[1432]) - \Omega(c[1234]) = 0$$
$$\Omega(c[1423]) - \Omega(c[1324]) = 0$$
$$\Omega(c[1243]) - \Omega(c[1342]) = 0$$

Kleiss-Kuijf Relations from Positive Geometry

[Damgaard, Ferro, Łukowski, Moerman 2103.13908]

Example: MHV_4

	(1234)	(1243)	(1324)	(1342)	(1423)	(1432)
$\langle 23\rangle \sim \alpha_4$	$+$	$+$	$-$	$-$	$+$	$-$
$\langle 24\rangle \sim \alpha_2+\alpha_4$	$+$	$+$	$+$	$-$	$-$	$-$
$\langle 34\rangle \sim \alpha_2$	$+$	$-$	$+$	$+$	$-$	$-$



Reflection symmetry relations:

$$A[1432] = A[1234]$$
$$A[1423] = A[1324]$$
$$A[1243] = A[1342]$$

equivalent positive geometries with *identical* canonical forms

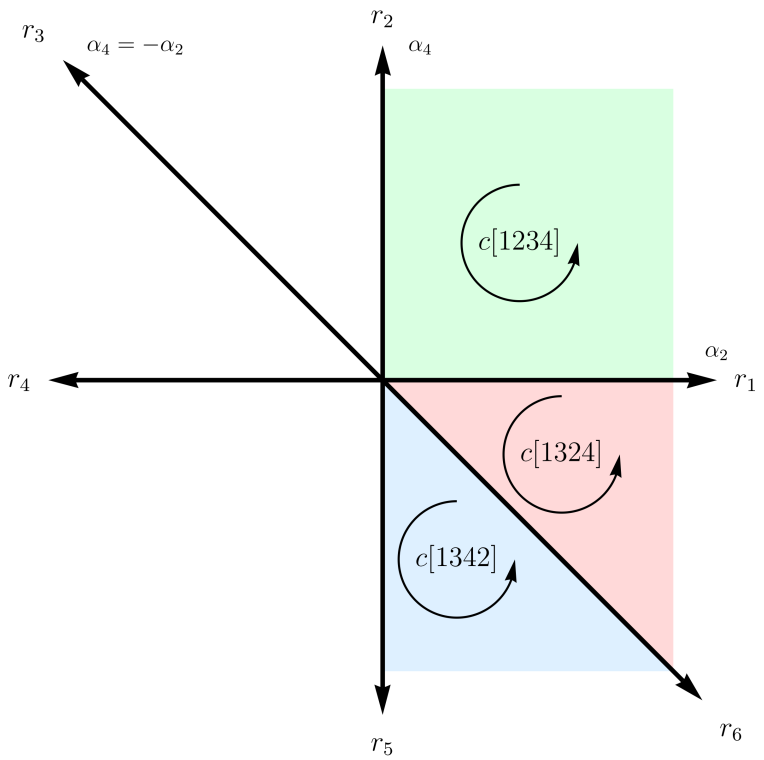
$$c[1432] \oplus c^-[1234]$$
$$c[1423] \oplus c^-[1324] \quad \text{have no vertices}$$
$$c[1243] \oplus c^-[1342]$$

Kleiss-Kuijf Relations from Positive Geometry

[Damgaard, Ferro, Łukowski, Moerman 2103.13908]

Example: MHV_4

	(1234)	(1243)	(1324)	(1342)	(1423)	(1432)
$\langle 23\rangle \sim \alpha_4$	$+$	$+$	$-$	$-$	$+$	$-$
$\langle 24\rangle \sim \alpha_2+\alpha_4$	$+$	$+$	$+$	$-$	$-$	$-$
$\langle 34\rangle \sim \alpha_2$	$+$	$-$	$+$	$+$	$-$	$-$



$U(1)$ decoupling relations:

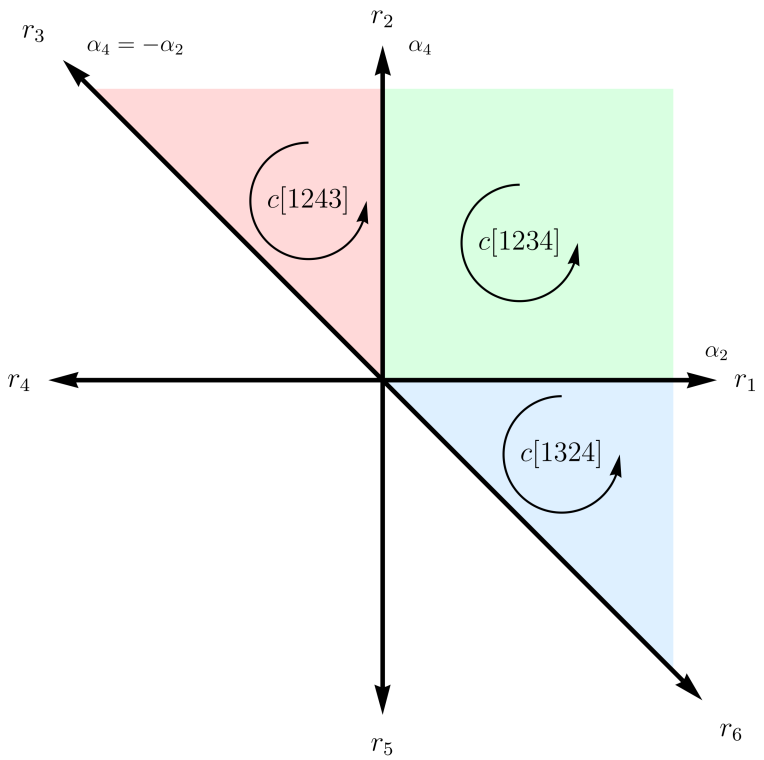
- $A[1234] + A[1324] + A[1342] = 0$
- $c[1234] \oplus c[1324] \oplus c[1342] = \mathbb{R}_{\geq 0} \times \mathbb{R}$ has no vertices

Kleiss-Kuijf Relations from Positive Geometry

[Damgaard, Ferro, Łukowski, Moerman 2103.13908]

Example: MHV_4

	(1234)	(1243)	(1324)	(1342)	(1423)	(1432)
$\langle 23\rangle \sim \alpha_4$	$+$	$+$	$-$	$-$	$+$	$-$
$\langle 24\rangle \sim \alpha_2+\alpha_4$	$+$	$+$	$+$	$-$	$-$	$-$
$\langle 34\rangle \sim \alpha_2$	$+$	$-$	$+$	$+$	$-$	$-$



$U(1)$ decoupling relations:

- $A[1243] + A[1234] + A[1324] = 0$
- $c[1243] \oplus c[1234] \oplus c[1324] \cong \mathbb{R}_{\geq 0} \times \mathbb{R}$ has no vertices

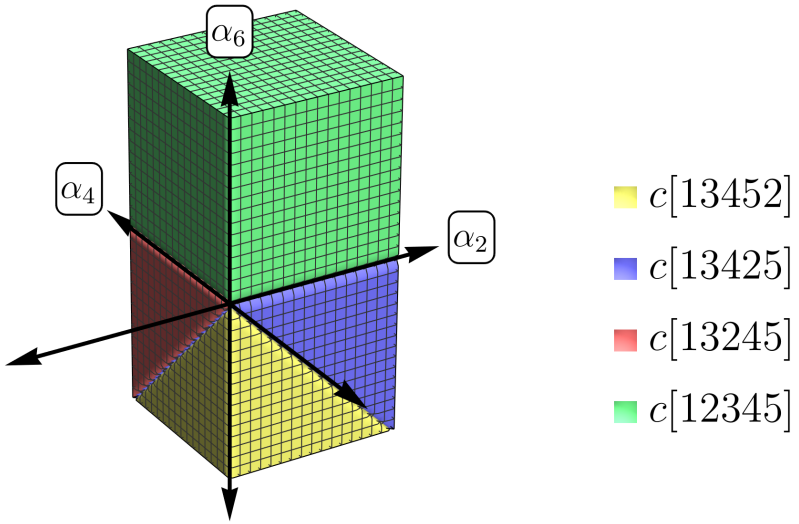
Kleiss-Kuijf Relations from Positive Geometry

[Damgaard, Ferro, Łukowski, Moerman 2103.13908]

Example: MHV_5

KK relations for $n = 5$: $A_5[1, \alpha, 5, \beta] = (-1)^{|\beta|} \sum_{\sigma \in \alpha \sqcup \beta^T} A_5[1, \sigma, 5]$

1. $|\beta| = 1$: $A[13452] = -A[13425] - A[13245] - A[12345]$



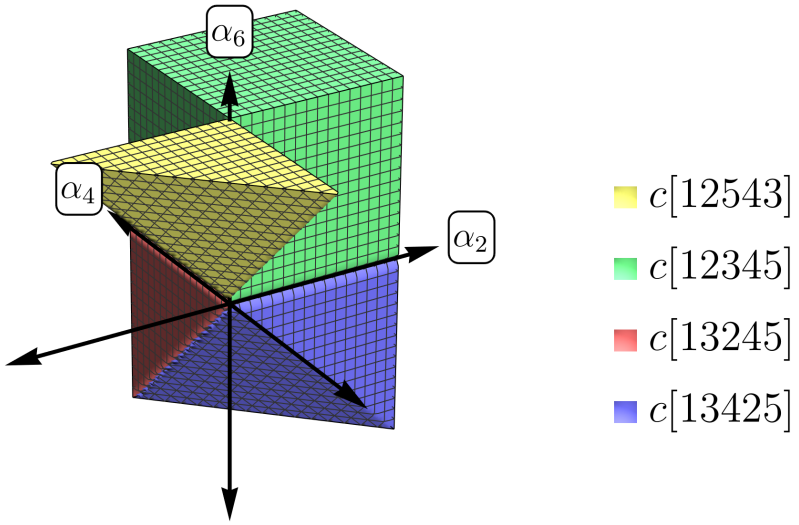
Kleiss-Kuijf Relations from Positive Geometry

[Damgaard, Ferro, Łukowski, Moerman 2103.13908]

Example: MHV_5

KK relations for $n = 5$: $A_5[1, \alpha, 5, \beta] = (-1)^{|\beta|} \sum_{\sigma \in \alpha \sqcup \beta^T} A_5[1, \sigma, 5]$

2. $|\beta| = 2$: $A[12543] = A[12345] + A[13245] + A[13425]$



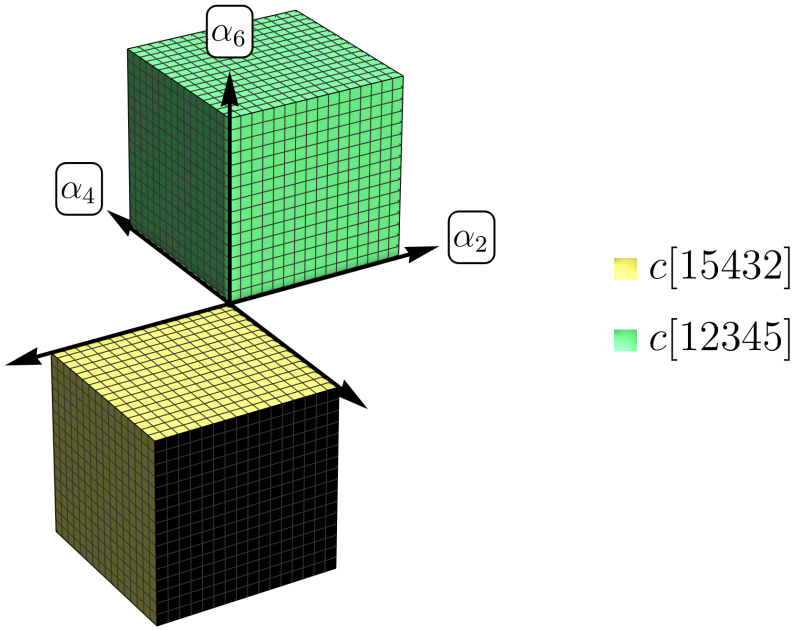
Kleiss-Kuijf Relations from Positive Geometry

[Damgaard, Ferro, Łukowski, Moerman 2103.13908]

Example: MHV_5

KK relations for $n = 5$: $A_5[1, \alpha, 5, \beta] = (-1)^{|\beta|} \sum_{\sigma \in \alpha \sqcup \beta^T} A_5[1, \sigma, 5]$

- $|\beta| = 3$: $A[15432] = -A[12345]$



Kleiss-Kuijf Relations from Positive Geometry

[Damgaard, Ferro, Łukowski, Moerman 2103.13908]

All Helicity Sectors: Poset-based Homological Algorithm

Input: boundary stratifications of $\mathcal{M}_{n,k}^{(\sigma)}$ for different σ

- momentum amplituhedra for different orderings *share the same vertices*
- sufficient to study momentum amplituhedra around a single vertex
- boundary stratifications for different orderings are *isomorphic*
- generate poset-intervals for different orderings using `amplituhedronBoundaries.m` [Łukowski, Moerman 2002.07146]

"pairs of rays pointing in opposite directions"

$\Longleftrightarrow$

one-dimensional boundaries inhabiting the same one-dimensional intersection of facet-defining hypersurfaces

Kleiss-Kuijf Relations from Positive Geometry

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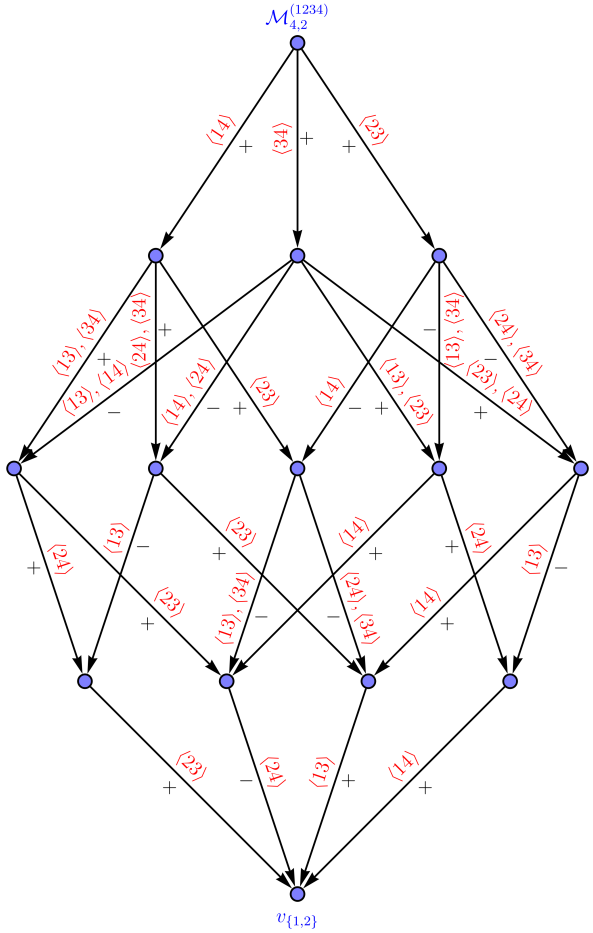
All Helicity Sectors: Poset-based Homological Algorithm

- draw Hasse diagrams of poset-intervals for each ordering around the same vertex
- label edges using boundary labels on slide 18
- decorate edges with signs such that every diamond multiplies to -1
- define boundary operators with respect to edge labels
- use boundary operators to find KK relations as oriented sums *without vertices*



Summary

Can we find a geometric origin for the Kleiss-Kuijf relations?

Yes: KK relations correspond to oriented sums *without vertices.*

$\Longrightarrow$ we can determine the Kleiss-Kuijf relations *directly* from geometry
(without needing the canonical form)

All Helicity Sectors: Poset-based Homological Algorithm

Input: boundary stratifications of $\mathcal{M}_{n,k}^{(\sigma)}$ for different σ

Outlook

- KK relations for other theories?
  - ✓ bi-adjoint ϕ^3 theory from the kinematic associahedron
  - • bi-adjoint ϕ^p theory ($p \geq 4$)?
- BCJ relations?
- formalize the notion of an oriented sum
- applications to other theories: eg. less-/non-supersymmetric theories, SUGRA, etc.?
- applications to loops?
- Loop momentum amplituhedron?
- ...

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Thank you!

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Kleiss-Kuijf Relations from Positive Geometry

[Damgaard, Ferro, Łukowski, Moerman 2103.13908]

MHV Amplitudes: Ray-based Homological Algorithm

- to each positive sector $c[\sigma] = \text{span}_{\mathbb{R}_{\geq 0}}\{r_{j_1^\sigma}, \ldots, r_{j_{n-2}^\sigma}\}$ assign a ray $(n-2)$ -form

$$\omega(c[\sigma]) = \frac{\det(r_{j_1^\sigma}, \ldots, r_{j_{n-2}^\sigma})}{|\det(r_{j_1^\sigma}, \ldots, r_{j_{n-2}^\sigma})|} \tilde{r}_{j_1^\sigma} \wedge \cdots \wedge \tilde{r}_{j_{n-2}^\sigma}$$

- define boundary operators with respect to facet-defining hyperplanes $h \in \mathcal{H}_n$

$$\partial_h(\tilde{r}_{j_1} \wedge \cdots \wedge \tilde{r}_{j_p}) = \omega(\text{span}_{\mathbb{R}_{\geq 0}}\{r_{j_1^\sigma}, \ldots, r_{j_p^\sigma}\} \cap h)$$

- define *graded vectorspace* $V_n \equiv \bigoplus_{i=0}^{n-2} V_n^{(i)}$ where

$$V_n^{(i)} = \text{span}_{\mathbb{Z}} \bigcap_{\{r_{j_1}, \ldots, r_{j_{n-2-i}}\} \in \binom{\mathcal{R}_n}{n-2-i}} \{\tilde{r}_{j_1} \wedge \cdots \wedge \tilde{r}_{j_{n-2-i}}\}$$

- each hyperplane $h \in \mathcal{H}_n$ induces

$$V_n^{(0)} \xrightarrow{\partial_h} V_n^{(1)} \xrightarrow{\partial_h} \cdots \xrightarrow{\partial_h} V_n^{(n-2)} = \mathbb{Z} \xrightarrow{\partial_h} 0$$

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MHV Amplitudes: Ray-based Homological Algorithm

- each $\{h_{i_1}, \ldots, h_{i_{n-2}}\} \in \binom{\mathcal{H}_n}{n-2}$ induces $V_n^{(0)} \xrightarrow{\partial_{h_{i_1}} \cdots \partial_{h_{i_{n-2}}}} V_n^{(n-2)} = \mathbb{Z}$

Claim: KK relations correspond to oriented sums *without vertices*.

KK relations for MHV amplitudes

$$\bigcap_{\{h_{i_1},\ldots,h_{i_{n-2}}\}\in\binom{\mathcal{H}_n}{n-2}} \ker\left(\partial_{h_{i_1}} \cdots \partial_{h_{i_{n-2}}}\right)$$

(checked for $n = 4, 5, 6, 7$)

Kleiss-Kuijf Relations from Positive Geometry

[Damgaard, Ferro, Łukowski, Moerman 2103.13908]

MHV Amplitudes: Ray-based Homological Algorithm

- each $\{h_{i_1}, \ldots, h_{i_{n-2}}\} \in \binom{\mathcal{H}_n}{n-2}$ induces $V_n^{(0)} \xrightarrow{\partial_{h_{i_1}} \cdots \partial_{h_{i_{n-2}}}} V_n^{(n-2)} = \mathbb{Z}$

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(checked for $n = 4, 5, 6, 7$)

Beyond MHV:

- simplifications no longer hold
- positivity conditions are non-linear $\implies$ "curvy" facets